

An Example: 2D Heat Equation

1 Model Problem

Consider the 2D heat equation on the space-time cylinder $Q := \Omega \times (0, T)$:

$$\partial_t u - \Delta u = f \quad \text{in } Q,$$

with homogeneous Dirichlet boundary conditions

$$u = 0 \quad \text{on } \partial\Omega \times (0, T),$$

and initial condition

$$u(x, 0) = u_0.$$

2 Weak Formulation

Multiply the PDE by a test function v and integrate over Ω :

$$\langle \partial_t u, v \rangle + \langle \nabla u, \nabla v \rangle = \langle f, v \rangle \quad \forall v \in H_0^1(\Omega).$$

In space-time form, define the spaces

$$U := \{u \in L^2(0, T; H_0^1(\Omega)) : \partial_t u \in L^2(0, T; H^{-1}(\Omega))\}, \quad V := L^2(0, T; H_0^1(\Omega)).$$

Then the weak problem can be written abstractly as: Find $u \in U$ such that

$$A(u)(v) = 0 \quad \forall v \in V,$$

where

$$A(u)(v) := \int_0^T \left(\langle \partial_t u, v \rangle + \langle \nabla u, \nabla v \rangle - \langle f, v \rangle \right) dt.$$

3 Space-time finite element discretization

Let $U_h \subset U$ and $V_h \subset V$ be finite-dimensional subspaces. The discrete primal problem is:

$$\text{Find } u_h \in U_h \text{ such that } A(u_h)(v_h) = 0 \quad \forall v_h \in V_h.$$

For the discrete adjoint, one solves

$$\text{Find } z_h \in V_h \text{ such that } A'(u_h)(v_h, z_h) = J'(u_h)(v_h) \quad \forall v_h \in U_h.$$

4 Goal functionals

Typical goal functionals include:

$$J(u) = \int_{\omega} u(x, T) dx,$$

where $\omega \subset \Omega$ is a target region.

5 DWR adjoint problem

The adjoint problem is defined by

$$A'(u)(v, z) = J'(u)(v) \quad \forall v \in U.$$

The unknown is the adjoint state z .

5.1 Gâteaux derivative

For a direction v , the Gâteaux derivative of A is

$$A'(u)(v, z) = \lim_{\varepsilon \rightarrow 0} \frac{A(u + \varepsilon v)(z) - A(u)(z)}{\varepsilon}.$$

For the heat equation this gives

$$A'(u)(v, z) = \int_0^T \left(\partial_t v, z + \nabla v, \nabla z \right) dt.$$

5.2 Integration by parts

Using integration by parts in time,

$$\int_0^T \partial_t v, z dt = v(T), z(T) - v(0), z(0) - \int_0^T v, \partial_t z dt.$$

Since the perturbation v preserves the initial condition, one has

$$v(0) = 0.$$

Using integration by parts in space,

$$\int_{\Omega} \nabla v, \nabla z dx = - \int_{\Omega} v \Delta z dx$$

provided $v = 0$ on $\partial\Omega$.

Hence the adjoint weak form becomes

$$v(T), z(T) + \int_0^T v, -\partial_t z - \Delta z dt = J'(u)(v).$$

6 Adjoint strong form

For the common choice

$$J(u) = \int_{\omega} u(x, T) dx,$$

we have

$$J'(u)(v) = \int_{\omega} v(x, T) dx = v(T), \mathbf{1}_{\omega}.$$

Therefore the adjoint strong form is

$$-\partial_t z - \Delta z = 0 \quad \text{in } \Omega \times (0, T),$$

$$z = 0 \quad \text{on } \partial\Omega \times (0, T),$$

$$z(\cdot, T) = \mathbf{1}_{\omega}.$$

This is a backward-in-time problem.

7 Error identity

Let $\tilde{u} \in U_h$ and $\tilde{z} \in V_h$ be arbitrary approximations. Then the DWR error identity is

$$J(u) - J(\tilde{u}) = \frac{1}{2} \left(\rho(\tilde{u})(z - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u - \tilde{u}) \right) - \rho(\tilde{u})(\tilde{z}) + R^{(3)},$$

where

$$\rho(\tilde{u})(\cdot) := -A(\tilde{u})(\cdot), \quad \rho^*(\tilde{u}, \tilde{z})(\cdot) := J'(\tilde{u})(\cdot) - A'(\tilde{u})(\cdot, \tilde{z}).$$

A third-order remainder term is

$$R^{(3)} = \frac{1}{2} \int_0^1 \left[J'''(\tilde{u} + se)(e, e, e) - A'''(\tilde{u} + se)(e, e, e, \tilde{z} + se^*) - 3A''(\tilde{u} + se)(e, e, e^*) \right] s(s-1) ds,$$

with

$$e = u - \tilde{u}, \quad e^* = z - \tilde{z}.$$

8 Enriched space approximation

Let $U_h^{(2)}$ and $V_h^{(2)}$ be enriched spaces with

$$U_h \subset U_h^{(2)} \subset U, \quad V_h \subset V_h^{(2)} \subset V.$$

The enriched primal and adjoint problems are

$$\text{Find } u_h^{(2)} \in U_h^{(2)} \text{ such that } A(u_h^{(2)})(v_h^{(2)}) = 0 \quad \forall v_h^{(2)} \in V_h^{(2)},$$

$$\text{Find } z_h^{(2)} \in V_h^{(2)} \text{ such that } A'(u_h^{(2)})(v_h^{(2)}, z_h^{(2)}) = J'(u_h^{(2)})(v_h^{(2)}) \quad \forall v_h^{(2)} \in U_h^{(2)}.$$

Then the computable estimator is

$$\eta^{(2)} = \frac{1}{2} \left(\rho(\tilde{u})(z_h^{(2)} - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u_h^{(2)} - \tilde{u}) \right) - \rho(\tilde{u})(\tilde{z}) + R_h^{(3)}.$$

A useful splitting is

$$\eta^{(2)} = \underbrace{\frac{1}{2} \left(\rho(\tilde{u})(z_h^{(2)} - \tilde{z}) + \rho^*(\tilde{u}, \tilde{z})(u_h^{(2)} - \tilde{u}) \right)}_{\eta_h^{(2)}} - \underbrace{\rho(\tilde{u})(\tilde{z})}_{\eta_k} + \underbrace{R_h^{(3)}}_{\eta_R^{(2)}}.$$